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In [1]: # K-Means Clustering + PCA using sklearn Wine Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score

# Step 1: Load Dataset
wine = load_wine()
df = pd.DataFrame(wine.data, columns=wine.feature_names)

# Step 2: Data Preprocessing (Scaling)
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df)

# Step 3: Elbow Method to find optimal clusters
wcss = []
for i in range(1, 11):
    kmeans_temp = KMeans(n_clusters=i, random_state=42)
    kmeans_temp.fit(scaled_data)
    wcss.append(kmeans_temp.inertia_)

plt.figure()
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()

# Step 4: Apply K-Means (k=3)
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(scaled_data)

df['Cluster'] = clusters

# Step 5: Evaluate Model
score = silhouette_score(scaled_data, clusters)
print("Silhouette Score:", score)

# Step 6: Apply PCA (2 components)
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

pca_df = pd.DataFrame(pca_data, columns=['PC1', 'PC2'])
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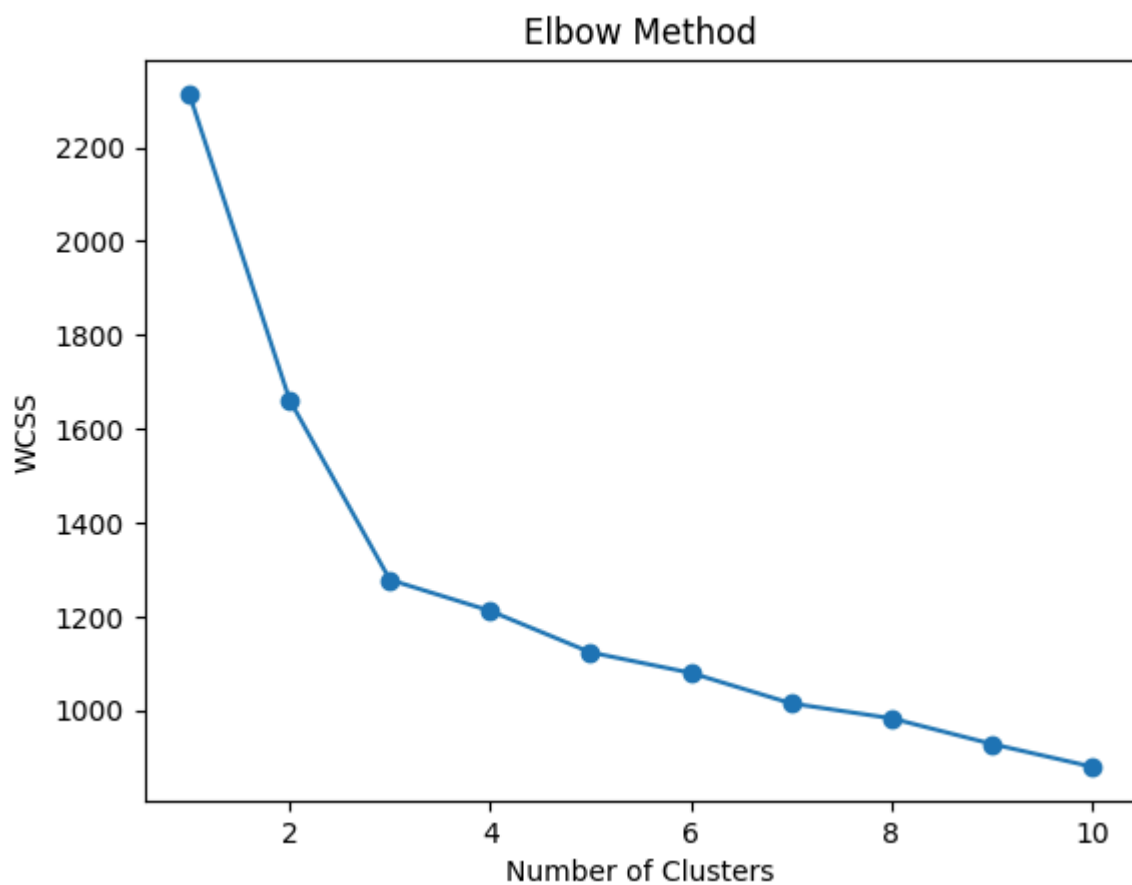
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pca_df['Cluster'] = clusters
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# Step 7: Visualization
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plt.figure()  
plt.scatter(pca_df['PC1'], pca_df['PC2'], c=pca_df['Cluster'])  
plt.xlabel("Principal Component 1")  
plt.ylabel("Principal Component 2")  
plt.title("K-Means Clusters (PCA Reduced Data)")  
plt.show()
```

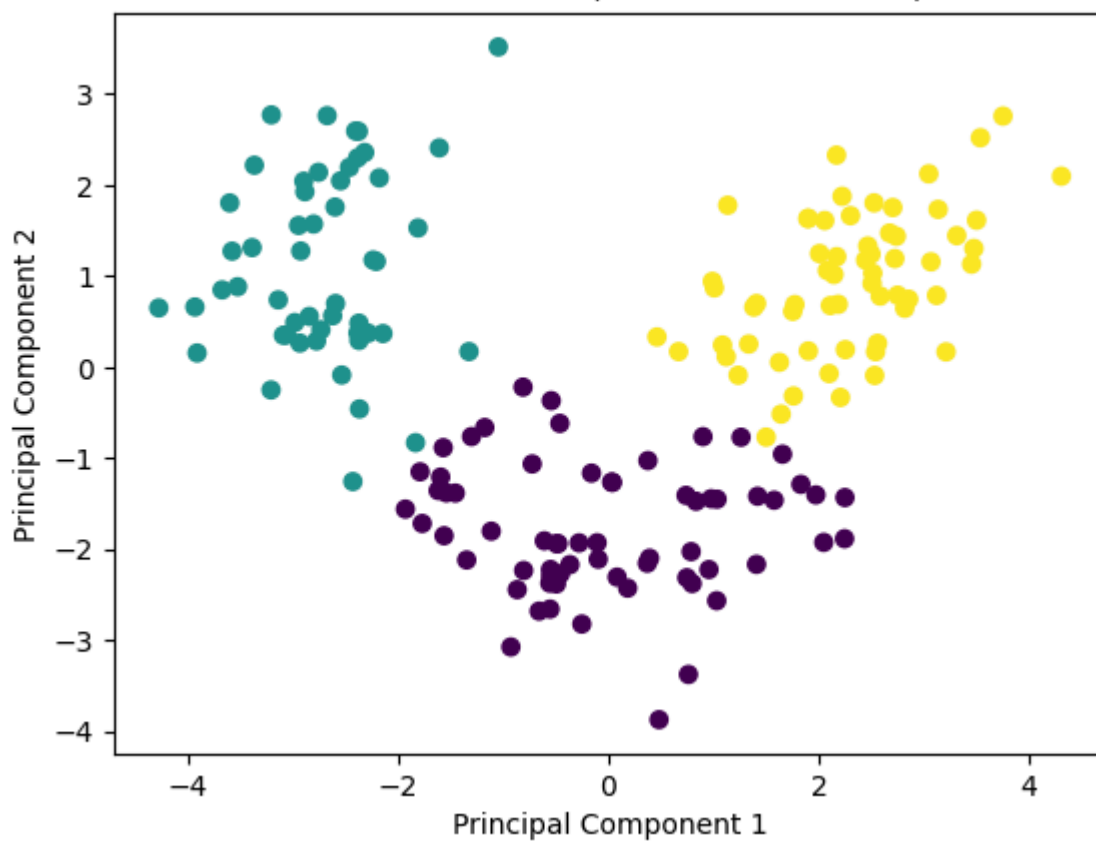
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# Step 8: Display Cluster-wise Mean (for business insight)
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cluster_summary = df.groupby('Cluster').mean()  
print("\nCluster-wise Feature Averages:\n")  
print(cluster_summary)
```



Silhouette Score: 0.2848589191898987

K-Means Clusters (PCA Reduced Data)



Cluster-wise Feature Averages:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium \
Cluster					
0	12.250923	1.897385	2.231231	20.063077	92.738462
1	13.134118	3.307255	2.417647	21.241176	98.666667
2	13.676774	1.997903	2.466290	17.462903	107.967742

	total_phenols	flavanoids	nonflavanoid_phenols	proanthocyanins \
Cluster				
0	2.247692	2.050000	0.357692	1.624154
1	1.683922	0.818824	0.451961	1.145882
2	2.847581	3.003226	0.292097	1.922097

	color_intensity	hue	od280/od315_of_diluted_wines	proline
Cluster				
0	2.973077	1.062708	2.803385	510.169231
1	7.234706	0.691961	1.696667	619.058824
2	5.453548	1.065484	3.163387	1100.225806